

PF Assignment #2

Group: Study Buddies

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Program 4: Online Shopping System

```
#include<iostream>

using namespace std;

string username, email, city, customerType, paymentMethod;

string products[8]={"T-Shirt","Jeans","Shoes","Watch","Handbag","Headphones","Mobile Cover","Perfume"};

int prices[8]={1200,3500,5000,2500,4200,3000,700,2800};

int cart[8]={0};

double productTotal;

void userRegister(){

    int choice, choice2;

    cout<<"==== Registration =====<<endl;

    cout<<"Username: ";

    getline(cin,username);

    cout<<"Email: ";

    cin>>email;

    cout<<"City: ";

    cin.ignore();

    getline(cin,city);

    do{

        cout<<"Customer Type: "<<endl;

        cout<<"1. New Customer"<<endl;

        cout<<"2. Returning Customer"<<endl;

        cout<<"\nChoose one of them (1or2): ";

        cin>>choice;
```

```

        if(choice==1){
            customerType="New Customer";
        }
        else if(choice==2){
            customerType="Returning Customer";
        }
        else{
            cout<<"\nInvalid Choice";
        }
    }while(choice!=1&&choice!=2);

    do{
        cout<<"Payment Method: "<<endl;
        cout<<"1. Debit/Credit Card"<<endl;
        cout<<"2. Cash on Delivery"<<endl;
        cout<<"\nChoose one of them (1or2): ";
        cin>>choice2;
        if(choice2==1){
            paymentMethod="Debit/Credit Card";
        }
        else if(choice2==2){
            paymentMethod="Cash on Delivery";
        }
        else{
            cout<<"\nInvalid Choice";
        }
    }while(choice2!=1&&choice2!=2);
}

void DisplayProducts(){
    cout<<"\n===== "<<endl;
    cout<<"Products List: "<<endl;
    for(int i=0;i<=7;i++){
        cout<<i+1<<"\t"<<products[i]<<"\t\t"<<prices[i]<<endl;
    }
}

```

```

    }

    cout<<"\n===== "<<endl;
}

```

```

void addProductsToCart(){
    DisplayProducts();

    int productNo, more, quantity;

    do{

        cout<<"\n\n Select Product (1 to 8):";

        cin>>productNo;

        cout<<"Enter Quantity: ";

        cin>>quantity;

        cart[productNo-1]+=quantity;

        cout<<"Product Added Successfully."<<endl;

        cout<<"Do you want to add more Products (1= Yes , 0= No) : ";

        cin>>more;

    }while(more==1);

    cout<<"\nItems added Successfully";

    cout<<"\n===== \n"<<endl;

}

```

```

void calculateProductTotal(){

    productTotal = 0;


    for(int i = 0; i < 8; i++){

        productTotal += cart[i] * prices[i];

    }

}

```

```

double calculateGST(){

    double gst;

    gst=productTotal*0.17;

    return gst;
}

```

```
}
```

```
int calculateDeliveryCharges(){  
    int deliveryCharges;  
    if(city=="Lahore" || city=="lahore" || city=="Karachi" || city=="karachi" || city=="Islamabad" || city=="islamabad"){  
        deliveryCharges=250;  
    }  
    else  
        deliveryCharges=500;  
  
    return deliveryCharges;  
}
```

```
double calculateCustomerDiscount(){  
    double customerDiscount;  
    if (customerType=="New Customer"){  
        customerDiscount=productTotal*0.05;  
    }  
    else  
        customerDiscount=productTotal*0.1;  
    return customerDiscount;  
}
```

```
double calculateOrderValueDiscount(){  
    double orderValueDiscount;  
    if(productTotal>=5000&&productTotal<=10000){  
        orderValueDiscount=productTotal*0.05;  
    }  
    else if(productTotal>10000){  
        orderValueDiscount=productTotal*0.12;  
    }  
    else  
        orderValueDiscount=0;  
    return orderValueDiscount;  
}
```

```
}
```

```
double calculatePaymentCharges(){
```

```
    double paymentCharges;
```

```
    if(paymentMethod=="Debit/Credit Card"){
```

```
        paymentCharges=productTotal*0.025;
```

```
    }
```

```
    else
```

```
        paymentCharges=0;
```

```
    return paymentCharges;
```

```
}
```

```
void displayCheckoutBill(){
```

```
    calculateProductTotal();
```

```
    double finalAmount;
```

```
    double gst=calculateGST();
```

```
    int deliveryCharges=calculateDeliveryCharges();
```

```
    double customerDiscount=calculateCustomerDiscount();
```

```
    double orderDiscount=calculateOrderValueDiscount();
```

```
    double paymentCharges=calculatePaymentCharges();
```

```
    finalAmount=productTotal+gst+deliveryCharges-customerDiscount-orderDiscount+paymentCharges;
```

```
    cout<<"===== ONLINE SHOPPING BILL ====="<<endl;
```

```
    cout<<"User Name: "<<username<<endl;
```

```
    cout<<"City: "<<city<<endl;
```

```
    cout<<"Customer Type: "<<customerType<<endl<<endl;
```

```
    cout<<"Product Total: Rs. "<<productTotal<<endl;
```

```
    cout<<"GST: Rs. "<<gst<<endl;
```

```
    cout<<"Delivery Charges: Rs. "<<deliveryCharges<<endl;
```

```
    cout<<"Customer Discount: Rs. "<<customerDiscount<<endl;
```

```
    cout<<"Order Discount: Rs. "<<orderDiscount<<endl;
```

```
    cout<<"Payment Charges: Rs. "<<paymentCharges<<endl<<endl;
```

```
    cout<<"-----"<<endl;
```

```

        cout<<"Final Payable Amount: Rs. "<<finalAmount<<endl;

        cout<<"Thank You for Shopping :)"<<endl;

        cout<<"======"<<endl;

    }

    void userDetails(){

        cout<<"\n===== USER DETAILS =====\n";

        cout<<"Name: "<<username<<endl;

        cout<<"Email: "<<email<<endl;

        cout<<"City: "<<city<<endl;

        cout<<"Customer Type: "<<customerType<<endl;

    }

```

```

int main(){

    userRegister();

    int choice;

    do{

        cout<<"\n=====Main Menu===== "<<endl;

        cout<<"1.      View Products"<<endl;

        cout<<"2.      Add Product to Cart "<<endl;

        cout<<"3.Calculate Checkout Bill"<<endl;

        cout<<"4.View User Details "<<endl;

        cout<<"5.Exit "<<endl;

        cout<<"\n Enter Choice: ";

        cin>>choice;

        if(choice==1){

```

```

        DisplayProducts();
    }
    else if(choice==2){
        addProductsToCart();
    }
    else if(choice==3){
        displayCheckoutBill();
    }
    else if(choice==4){
        userDetails();
    }
    else if(choice==5){
        cout<<"Program Exiting...";
    }
    else
        cout<<"\nInvalid Choice!"<<endl;
    }while(choice!=5);
}

```

Program 6: Personal Student Diary

```

#include<iostream>

using namespace std;

string username, password;

string titles[20];
string contents[20];
int entry=0;

void userRegister(){
    cout<<"===== Register =====\n"<<endl;
    cout<<"Username: ";
    //cin.ignore();
}

```

```

        getline(cin,username);

        cout<<"Password: ";

        getline(cin,password);

        cout<<"Register Successfull"<<endl;

    }

    bool userLogin(){

        string loginUsername, loginPassword;

        cout<<"\n===== Login =====\n"<<endl;

        cout<<"Username: ";

        getline(cin,loginUsername);

        cout<<"Password: ";

        getline(cin,loginPassword);

        if((loginUsername==username&&loginPassword==password)

        {

            cout<<"\nLogin Successfull\n";

            return true;

        }

        else{

            cout<<"\nInvalid Username or Password\n";

            return false;

        }

    }

    void createNewEntry()

    {

        string title, content;

        if (entry>20){

            cout<<"\nYou can store only 20 Diaries!"<<endl;

        }

        else{

            do{

                cout<<"\nTitle: ";

                getline(cin,title);

                if(title.empty()){

```

```

        cout<<"\nTitle cannot be empty!\n";
    }
}while(title.empty());

do{
    cout<<"Content: ";
    getline(cin,content);
    if(content.empty()){
        cout<<"\nContent cannot be empty!\n";
    }
}
while(content.empty());

titles[entry]=title;
contents[entry]=content;
entry++;
}
}

void viewEntries()
{
    cout<<"\n=====View Entries=====\\n";
    for(int i=0; i<entry;i++){
        cout<<endl<<i+1<<" ". "<<titles[i]<<endl;
    }
    cout<<"Total Entries: "<<entry<<endl<<endl;
}

void readEntries()
{
    string title;
    cout<<"\n===== Read Entries =====\\n";
    for(int i=0; i<entry;i++){
        cout<<endl<<i+1<<" ". "<<titles[i]<<endl;
    }
    cout<<"Total Entries: "<<entry<<endl<<endl;
}

```

```

        cout<<"\nEnter the title that you want to see Entry: ";

        getline(cin,title);

        for(int i=0;i<entry; i++){

            if(title==titles[i]){

                cout<<"Title: "<<titles[i]<<endl;

                cout<<"Content: "<<contents[i]<<endl;

            }

        }

    }

}

void editEntry()

{

    cout<<"\n=====\\n";

    cout<<"      Edit Entry      \\n";

    cout<<"=====\\n";

    string title;

    for(int i=0; i<entry;i++){

        cout<<endl<<i+1<<" ".<<titles[i]<<endl;

    }

    cout<<"Total Entries: "<<entry<<endl<<endl;

    cout<<"\nEnter the title that you want to edit Entry: ";

    getline(cin,title);

    for(int i=0;i<entry; i++){

        if(title==titles[i]){

            cout<<"New Title: ";

            getline(cin,titles[i]);

            cout<<"New Content: ";

            getline(cin,contents[i]);

        }

    }

    cout<<"\nEntry edit successfully!"<<endl;

}

void deleteEntry()

{

```

```

string title;

int index;

cout<<"\n=====\\n";

cout<<"      Delete Entry      \\n";

cout<<"=====\\n";

for(int i=0; i<entry;i++){

    cout<<endl<<i+1<<" . "<<titles[i]<<endl;

}

cout<<"Total Entries: "<<entry<<endl<<endl;

cout<<"Enter the title of entry that you want delete: ";

getline(cin,title);

for(int i=0;i<entry;i++){

    if(title==titles[i]){

        index=i;

    }

}

for(int j=index;j<entry;j++){

    titles[j] = titles[j+1];

contents[j] = contents[j+1];

}

entry--;

cout<<"Entry deleted successfully!"<<endl;

}

void DeleteAllEntry()

{

    if(entry==0){

        cout<<"The Entries are already 0"<<endl;

    }

    else{

        entry=0;

        cout<<"All entries deleted successfully!"<<endl;

    }

}

```

```
}
```

```
int main(){  
    userRegister();  
    bool loggedin=false;  
    while (!loggedin){  
        loggedin=userLogin();  
    }  
    int choice;  
    do{  
        cout<<"===== "<<endl;  
        cout<<"      Menu      "<<endl;  
        cout<<"===== "<<endl;  
        cout<<"\n1. Create New Diary Entry"<<endl;  
        cout<<"2. View All Entries"<<endl;  
        cout<<"3. Read Full Entry"<<endl;  
        cout<<"4. Edit an Entry"<<endl;  
        cout<<"5. Delete One Entry"<<endl;  
        cout<<"6. Delete All Entries"<<endl;  
        cout<<"7. Exit"<<endl;  
        cout<<"\nEnter Your Choice: ";  
        cin>>choice;  
        cin.ignore();  
        if(choice==1){  
            createNewEntry();  
        }  
        else if(choice==2){  
            viewEntries();  
        }  
        else if(choice==3){  
            readEntries();  
        }  
        else if(choice==4){
```

```

        editEntry();
    }
    else if(choice==5){
        deleteEntry();
    }
    else if(choice==6){
        DeleteAllEntry();
    }
}
while(choice!=7);
}

```

Program 5: Social Media Management System

```
#include<iostream>
```

```

using namespace std;

string platforms[3]={"Facebook", "Instagram", "LinkedIn"};
double platformCharges[3]={12000, 15000, 20000};
string clientName, Business, businessType;
int duration;
string Platform;
int baseCharges;
int totalBill;
int postCost=1000, reelCost=2500, carouselCost=1800;
int adBudget, designCharges, extradurationcharges;
double handlingFee, GST, serviceCost,discount;

int registerClient(){
    int choice;
    cout<<"Enter Name: ";
    cin.ignore();
    getline(cin, clientName);
    cout<<"Business Name: ";

```

```

getline(cin, Business);

cout<<"Business Type: \n1. Small business\n2. Medium business\n3. Corporate business"<<endl;

cout<<"Choice: ";

cin>>choice;

if (choice==1){

    businessType="Small Business";

}

else if(choice==2){

    businessType="Medium business";

}

else if(choice==3){

    businessType="Corporate Business";

}

else

cout<<"Invalid Input Program terminate";


cout<<"Enter Dubration(Days): ";

cin>>duration;

cout<<"Client Register Successfully\n\n";

}

int extraDuration(){

    if (duration>30){

        int extraDays;

        extraDays=duration-30;

        extradurationcharges=extraDays*500;

        return extradurationcharges;

    }

    else

        return 0;

}

int selectPlatform(){

    cout<<"=====Platforms===== "<<endl;

    cout<<"No.    Platform    Base Management Charges"<<endl;

```

```

        cout<<"1."<<platforms[0]<<"\t"<<platformCharges[0]<<endl;
        cout<<"2."<<platforms[1]<<"\t"<<platformCharges[1]<<endl;
        cout<<"3."<<platforms[2]<<"\t"<<platformCharges[2]<<endl;
        cout<<"Chose one of them(1-3): ";

        int select;

        cin>>select;

if(select==1){
        Platform=platforms[0];
        baseCharges=platformCharges[0];
    }
    else if(select==1){
        Platform=platforms[1];
        baseCharges=platformCharges[1];
    }
    else if(select==1){
        Platform=platforms[2];
        baseCharges=platformCharges[2];
    }
    else
        cout<<"Invalid Input Program Terminate";

        cout<<"Platform Select Successfully\n\n";
}

void calculateCost(){
        int post=0, reel=0, carousel=0;

        cout<<"Number of static post: ";
        cin>>post;

        cout<<"Number of reel/videos: ";
        cin>>reel;

        cout<<"Number of Carousel Post: ";
        cin>>carousel;

        cout<<"Enter ad budget: ";
        cin>>adBudget;

```

```

designCharges=(post*postCost)+(reel*reelCost)+(carousel*carouselCost);

}

int adHandlingFee(){
    if(adBudget<50000){
        handlingFee=(adBudget*5)/100;
    }
    else if(adBudget>=50000&&adBudget<=100000){
        handlingFee=(adBudget*8)/100;
    }
    else if(adBudget>100000){
        handlingFee=(adBudget*10)/100;
    }
    else
        handlingFee=0;
}

int totalServiceCost(){
    serviceCost=baseCharges+designCharges+handlingFee+extradurationcharges;
}

int gst(){
    GST=(serviceCost*16)/100;
}

int calculatediscount(){
    if(businessType=="Small Business"){
        discount=(serviceCost*5)/100;
    }
    else if(businessType=="Medium business"){
        discount=(serviceCost*8)/100;
    }
    else{
        discount=(serviceCost*10)/100;
    }
}

```

```

}

void finalBill(){
    calculateCost();
    adHandlingFee();
    totalServiceCost();
    extraDuration();
    gst();
    calculatediscount();

    cout<<"===== SOCIAL MEDIA CAMPAIGN BILL =====\n"<<endl;

    cout<<"Client Name: "<<clientName<<endl;
    cout<<"Business Name: "<<Business<<endl;
    cout<<"Business Type: "<<businessType<<endl;
    cout<<"Selected Platform: "<<Platform<<endl;
    cout<<"Campaign Duration: "<<duration<<" Days"<<endl;

    cout<<"Platform Management Charges: Rs. "<<baseCharges<<endl;
    cout<<"Post Design Cost: Rs. "<<designCharges<<endl;
    cout<<"Ad Budget: Rs. "<<adBudget<<endl;
    cout<<"Ad Handling Fee: Rs. "<<handlingFee<<endl;
    cout<<"Extra Duration Charges: Rs. "<<extradurationcharges<<endl;
    cout<<"GST: Rs. "<<GST<<endl;
    cout<<"Discount: Rs. "<<discount<<endl;
    totalBill=baseCharges+designCharges+adBudget+handlingFee+extradurationcharges+GST-discount;
    cout<<"-----"<<endl;
    cout<<"Final Campaign Cost: Rs. "<<totalBill<<endl;
    cout<<"===== "<<endl;

}

int viewClient(){
    cout<<"Client Name: "<<clientName<<endl;
    cout<<"Business Name: "<<Business<<endl;
    cout<<"Business Type: "<<businessType<<endl;

```

```

        cout<<"Campaign Duration: "<<duration<<" Days"<<endl;

        cout<<"Selected Platform: "<<Platform<<endl;
    }

    void viewPlatform(){

        cout<<"=====Platforms===== "<<endl;

        cout<<"No.   Platform   Base Management Charges"<<endl;

        cout<<"1."<<platforms[0]<<"\t"<<platformCharges[0]<<endl;
        cout<<"2."<<platforms[1]<<"\t"<<platformCharges[1]<<endl;
        cout<<"3."<<platforms[2]<<"\t"<<platformCharges[2]<<endl;

    }

    int main()
    {

        int choice;

        registerClient();

        do{

            cout<<"=====Main Menu===== "<<endl;

            cout<<"1. View Platforms"<<endl;

            cout<<"2. Select Campaign Platform"<<endl;

            cout<<"3. Calculate Campaign Cost "<<endl;

            cout<<"4. View Client Details "<<endl;

            cout<<"5. Exit "<<endl;

            cout<<"Enter Choice: ";

            cin>>choice;

            if(choice==1){

                viewPlatform();

            }

            else if(choice==2){

                selectPlatform();

            }

            else if(choice==3){

                finalBill();

            }

            else if(choice==4){

```

```

        viewClient();
    }
}

while(choice!=5);
}

```

Program 2: Restaurant Billing Management System

```

#include<iostream>

#include<string>

using namespace std;

    string name ;

    string contact_number;

    int person_count;

    int order_type;

void registration(){

    cout<< " \t =====coustomer Registration===== "<< endl;

    cout<< " \t Enter your name : ";

    cin>>name ;

    cout<<" \t Enter your number : ";

    cin>> contact_number ;

    cout<< " \t choose\n \t1. DINE IN\n \t2. TAKEAWAY ";

    cout<< " \t enter your choice : ";

    cin>> order_type ;

    if(order_type== 1 ){

        cout<< " \t Dine in "<< endl;

        cout<< " \t Enter the person count :";

        cin>> person_count;

        cout<< " \t=====COUSTOMER REGISTERED=====\\n";

        cout<<"\t-----\\n";

    }

    else if ( order_type == 2){

```

```

        cout<< " \t Takeaway "<< endl;

        cout<< " \t Enter the person count :";

        cin>> person_count;

        cout<< " \t=====COUSTOMER REGISTERED=====\\n";

        cout<<"\t-----\\n";
    }

    else{
        cout<< " \t invalid "<< endl;

        cout<<"\t-----\\n";
    }
}

string menu[8]= {

    " Chicken Burger  ",
    " Zinger Burger  ",
    " Pizza Small    ",
    " Pizza Large    ",
    " Chicken Biryani ",
    " BBQ Platter    ",
    " Fries          ",
    " Cold Drink     "

};

int price[8] = {

    450,
    550,
    900,
    1800,
    350,
    1200,
    250,
    120

};

int total = 0;

```

```

void show_menu(){

    cout<< "\n          =====MENU===== "<<endl;

    for(int i = 0; i < 8; i++) {

        cout << i + 1 << " . " << menu[i] << "\t Rs : " << price[i] <<endl;

    }

}

void selection(){

    cout<< "\t --PLACE YOUR ORDER-- " <<endl;

    cout<< "\t -- AFTER SELECTION PRESS 0 -- " <<endl;

    int choice ;

    do{

        cout<< "\t ADD : ";

        cin>> choice ;

        if ( choice>=1 && choice<=8){

            cout<< "\t. "<< menu[ choice -1 ]<< price [ choice - 1] <<endl;

            total = total + price[choice - 1 ];

        }

    }while( choice!=0);

}

int total_bill(){

    cout<< "\t Your total food bill : "<< total << endl;

    return total;

}

float service(){

    float charges = 0;

    if( order_type == 1)

        charges = total * 0.10 ;

    else if( order_type == 2 )

        charges = total * 0.05;

}

```

```

        cout<< " \t service charges bill : "<< charges << endl;

        return charges;
    }

float gst(){

    float tax = 0;

    tax = total * 0.16;

    cout<< " \t gst : "<< tax << endl;

    return tax ;
}

float discount(){

    float discount = 0 ;

    if ( total >= 3000 && total <= 5000){

        discount = total * 0.05;

    }

    else if ( total >= 5001 && total <= 10000){

        discount = total * 0.10 ;

    }

    else if ( total > 10000){

        discount = total * 0.15;

    }

    cout<< " \t discount : "<< discount << endl;

    return discount ;
}

float final_bill(){

    float final_bill = 0 ;

    return total_bill () + service() + gst() - discount();

}

void delivery(){

    if(final_bill() > 5000){

        cout << "\nFree Delivery Available!" << endl;

    }
}

```

```

}

void cust_detail(){

    cout<<" \n    =====CUSTOMER DETAIL===== "<< endl;

    cout<<" \t Customer Name : "<< name << endl;

    cout<<" \t Contact Number : "<< contact_number<<endl;

    cout<<" \t Person count : "<< person_count<<endl;

    cout<<" \t Order type : "<< order_type<<endl;

    cout << "\t Order type : ";

}

if(order_type == 1){

    cout << "\tDine In" << endl;

}

else if(order_type == 2){

    cout << "\tTakeaway" << endl;

}

cout<<"\t-----"<<endl;

}

int main(){

    int option ;

    registration();

    do{

        cout<<" \t====MAIN MENU===="<<endl;

        cout << "\t 1. View Food Menu\n";

        cout << "\t 2. Place Order\n";

        cout << "\t 3. Calculate Bill\n";

        cout << "\t 4. View Customer Details\n";

        cout << "\t 5. Exit\n";

        cout<<" \t Enter your choice : ";

        cin>> option ;

        switch(option){

            case 1:

                show_menu();

                cout<<"\t-----\n";

```

```

break;

    case 2:

show_menu();

selection();

cout<<"\t-----\n";

break;

    case 3:

cout<<"\t FINAL BILL : "<<final_bill();

cout<<"\n\t-----\n";

break;

    case 4:

cust_detail();

break;

    case 5:

cout << "\n    Thank You For Visiting!  \n";

break;


    default:

cout << "\nInvalid Choice!\n";

    }

}while( option !=5);

}

```

Program 3: Grocery Mart Billing System

```

#include <iostream>

using namespace std ;

string Name,id ,CustomerType ,PaymentMethod ;

string itemNames[8]={"Rice 1 KG", "Sugar 1 KG", "CookingOil 1L", "Milk Pack", "Tea Pack", "Flour 5 KG", "Eggs Dozen",
"Detergent"};

float itemPrices[8]={350, 180, 580, 220, 450, 950, 320, 600};

int cart[8]={0};

void CustomerRegistration(){

    cout<<"\t----- * CUSTOMER REGISTRAION * -----" << endl;

```

```

        cout<<" \t Customer name :";

        cin>>Name;

        cout<<" \t Customer id #:";

        cin>>id;

        cout<<" Customer Type \n \t1. regular \n \t2. member: ";

        int typeChoice;

cin>>typeChoice;

if (typeChoice ==1)

    CustomerType ="Regular customer";

else if (typeChoice== 2)

    CustomerType ="Membership customer";

else

    cout<<"Invalid choice!"<<endl;

        cout<<" \t PaymentMethod \n \t1. cash \n \t2. card :";

        int payChoice;

cin>>payChoice;

if (payChoice == 1)

    PaymentMethod = "Cash";

else if (payChoice == 2)

    PaymentMethod = "Card";

else

    cout << "Invalid choice!" << endl;

cout<<" \t Customer Type: " << CustomerType << endl;

cout<<" \t Payment Method: " << PaymentMethod << endl;

cout<<" \t-----***-----"<<endl ;

}

void displayGroceryList(){

    cout<<"\t----- * Grocery Items * -----"<<endl;

    cout<<"item no.\t \t Grocery Items \t\t price"<<endl;

    for(int i = 0; i < 8; i++) {

        cout<< i+1 << ".\t\t\t" << itemNames[i] << "\t\tRs. " << itemPrices[i] << endl;

    }
}

```

```
    cout<<"\t-----***-----"<<endl;
}
```

```
void displayMenu(){
    cout<<"\n\t-----MAIN MENU-----"<<endl;
    cout<<"1. View Grocery Items"<<endl;
    cout<<"2. Add Items to Cart"<<endl;
    cout<<"3. Calculate Bill"<<endl;
    cout<<"4. View Customer Details"<<endl;
    cout<<"5. Exit" << endl;
    cout<<"Enter your choice: ";
}
```

```
void addItemToCart() {
    int itemNo, qty, more;
    displayGroceryList();
    do {
        cout<<"\nEnter Item Number (1-8): ";
        cin>> itemNo;
        cout<<"Enter Quantity: ";
        cin>> qty;
        cart[itemNo - 1] += qty;
        cout<< itemNames[itemNo-1]<<"added!"<<endl;
        cout<<"Add more items? (1=Yes, 2=No): ";
        cin>> more;
    } while(more == 1);
    cout<<"Items added successfully!"<< endl;
}
```

```
void calculateBill() {
    float grossBill = 0;
    for(int i = 0; i < 8; i++) {
        grossBill += itemPrices[i] * cart[i];
    }
}
```

```

}

float foodBill = 0, nonFoodBill = 0;

for(int i = 0; i < 7; i++) {
    foodBill += itemPrices[i] * cart[i];
}

nonFoodBill = itemPrices[7] * cart[7];

float salesTax = (foodBill * 0.05) + (nonFoodBill * 0.10);

float memberDiscount = 0;

if(CustomerType == "Membership customer")
    memberDiscount = grossBill * 0.07;

float billDiscount = 0;

if(grossBill >= 5000 && grossBill <= 10000)
    billDiscount = grossBill * 0.05;
else if(grossBill > 10000)
    billDiscount = grossBill * 0.10;

float totalBill = grossBill + salesTax - memberDiscount - billDiscount;

float cardCharges = 0;

if(PaymentMethod == "Card")
    cardCharges = totalBill * 0.02;

float finalBill = totalBill + cardCharges;

float loyaltyPoints = finalBill / 100;

cout << "\t Generated Loyalty Points: "<<loyaltyPoints<<endl;

float oldPoints;

cout << "\t Enter Existing Points (0 if new): ";

cin >> oldPoints;

float totalPoints = loyaltyPoints + oldPoints;

cout << "\t Total Loyalty Points: "<< totalPoints << endl;

int redeem;

cout << "\t Press 1 to Redeem, Press 2 to Skip: ";

cin >> redeem;

if(redeem == 1)
    finalBill = finalBill - totalPoints;

cout << "\t-----***-----"<<endl;

```

```

cout << "\t Gross Bill    : Rs." << grossBill << endl;

cout << "\t Sales Tax     : Rs." << salesTax << endl;

cout << "\t Member Discount: Rs." << memberDiscount << endl;

cout << "\t Bill Discount  : Rs." << billDiscount << endl;

cout << "\t Card Charges   : Rs." << cardCharges << endl;

cout << "\t Loyalty Points : " << totalPoints << endl;

cout << "\t-----" << endl;

cout << "\t Final Bill : Rs. " << finalBill << endl;

cout << "\t Thank You For Shopping :)" << endl;

cout << "\t-----***-----" << endl;
}

```

```

void displayCustomerDetails() {

    cout << "\t----- * CUSTOMER DETAILS* -----" << endl;

    cout << "\t Customer Name : " << Name << endl;

    cout << "\t Customer ID : " << id << endl;

    cout << "\t Customer Type : " << CustomerType << endl;

    cout << "\t Payment Method: " << PaymentMethod << endl;

    cout << "\t-----***-----" << endl;
}

```

```

int main () {

    CustomerRegistration ();

    int choice;

    do {

        displayMenu();

        cin >> choice;

        if (choice == 1) displayGroceryList();

        else if (choice == 2) addItemsToCart();

        else if (choice == 3) calculateBill();

        else if (choice == 4) displayCustomerDetails();

        else if (choice == 5) cout << "Goodbye!" << endl;

        else cout << "Invalid choice!" << endl;
    }
}

```

```
    } while(choice != 5);  
  
    return 0;  
  
}
```

Program 1: Electricity Bill Management System for LESCO

```
#include <iostream>  
  
#include <string>  
  
using namespace std;  
  
// -----GLOBAL Variables  
  
string customerName;  
  
string customerType;  
  
int customerID;  
  
int meterOption;  
  
double slabRates[8] = {12.21, 14.53, 31.51, 38.41,41.62, 43.04, 44.18, 49.10};  
  
// Function prototype  
  
void registerCustomer();  
  
void displayMenu();  
  
double calculateConsumptionCharges(int units);  
  
double calculateGST(double amount);  
  
double calculateIncomeTax(double amount);  
  
double calculateElectricityDuty(double consumptionCharges);  
  
double calculateFixedCharges(int units);  
  
double calculateNewConnectionCharges();  
  
void displayBill();  
  
void displayCustomerDetails();  
  
// Main function  
  
int main(){  
  
    registerCustomer();
```

```

    displayMenu();

    return 0;
}

// Customer Register
void registerCustomer(){

    cout<< "===== CUSTOMER REGISTRATION ====="<<endl;

    cout<< "Enter Customer Name: ";

    cin.ignore();

    getline(cin, customerName);

    cout<< "Enter Customer ID: ";

    cin>> customerID;

    int typeChoice;

    cout<< "\nCustomer Type\n";

    cout<< "1. Household\n";

    cout<< "2. Commercial\n";

    cout<< "Enter Choice: ";

    cin>> typeChoice;

    if (typeChoice == 1)

        customerType = "Household";

    else

        customerType = "Commercial";

    cout<< "\nMeter Option\n";

    cout<< "1. First Meter\n";

    cout<< "2. Second Meter\n";

    cout<< "Enter Choice: ";

    cin>> meterOption;

}

// Display Menu
void displayMenu()

{

```

```
int choice;

do
{
    cout<< "\n===== LESCO MENU =====\n";
    cout<< "1. Calculate Monthly Bill\n";
    cout<< "2. Apply For New Connection\n";
    cout<< "3. View Customer Details\n";
    cout<< "4. Exit\n";

    cout<< "Enter Choice: ";
    cin>> choice;

    switch (choice)
    {
        case 1:
            displayBill();
            break;

        case 2:
            cout << "\nNew Connection Charges = Rs. "
                << calculateNewConnectionCharges() << endl;
            break;

        case 3:
            displayCustomerDetails();
            break;

        case 4:
            cout << "\nProgram Exited Successfully.\n";
            break;

        default:
```

```

        cout << "\nInvalid Choice!\n";
    }

} while (choice != 4);
}

// Consumpton Charges

double calculateConsumptionCharges(int units)
{
    double rate;

    if (units <= 100)
        rate = slabRates[0];
    else if (units <= 200)
        rate = slabRates[1];
    else if (units <= 300)
        rate = slabRates[2];
    else if (units <= 400)
        rate = slabRates[3];
    else if (units <= 500)
        rate = slabRates[4];
    else if (units <= 600)
        rate = slabRates[5];
    else if (units <= 700)
        rate = slabRates[6];
    else
        rate = slabRates[7];

    return units * rate;
}

// calculate Gst

```

```
double calculateGST(double amount)
```

```
{  
    return amount * 0.18;  
}
```

```
// income tex
```

```
double calculateIncomeTax(double amount)
```

```
{  
    if (customerType == "Household")  
        return amount * 0.10;  
    else  
        return amount * 0.15;  
}
```

```
// Electricity duty
```

```
double calculateElectricityDuty(double consumptionCharges)
```

```
{  
    return consumptionCharges * 0.015;  
}
```

```
// Fixed charges
```

```
double calculateFixedCharges(int units)
```

```
{  
    if (units <= 300)  
        return 0;  
    else if (units <= 400)  
        return 200;  
    else if (units <= 500)  
        return 400;
```

```

else if (units <= 600)
    return 600;
else if (units <= 700)
    return 800;
else
    return 1000;
}

// New connection

double calculateNewConnectionCharges()
{
    double meterCost = 10000;
    double cableCost = 5000;
    double securityCost = 15000;
    double additionalCharges = 0;

    if (customerType == "Household")
    {
        if (meterOption == 1)
            additionalCharges = 2500;
        else
            additionalCharges = 5000;
    }
    else
    {
        if (meterOption == 1)
            additionalCharges = 35000;
        else
            additionalCharges = 70000;
    }

    double properConnectionCost = 250000;

```

```
    return meterCost + cableCost + securityCost +  
        additionalCharges + properConnectionCost;  
}
```

```
// Display Bill
```

```
void displayBill()
```

```
{
```

```
    int units;
```

```
    cout<< "\nEnter Units Consumed: ";
```

```
    cin>> units;
```

```
    double consumptionCharges =
```

```
        calculateConsumptionCharges(units);
```

```
    double electricityDuty =
```

```
        calculateElectricityDuty(consumptionCharges);
```

```
    double fixedCharges =
```

```
        calculateFixedCharges(units);
```

```
    double meterRent = 250;
```

```
    double tvFee = 35;
```

```
    double subTotal = consumptionCharges + electricityDuty + fixedCharges + meterRent + tvFee;
```

```
    double gst =
```

```
        calculateGST(subTotal);
```

```
    double incomeTax =
```

```
        calculateIncomeTax(subTotal);
```

```

double totalBill =subTotal +gst +incomeTax;

cout<< "\n===== LESCO ELECTRICITY BILL =====\n";

cout<< "Customer Name : " << customerName << endl;
cout<< "Customer ID  : " << customerID << endl;
cout<< "Customer Type : " << customerType << endl;
cout<< "Units Consumed: " << units << endl;

cout<< "\nElectricity Consumption Charges: Rs. "<< consumptionCharges << endl;
cout<< "Electricity Duty: Rs. "<< electricityDuty << endl;
cout<< "Fixed Charges: Rs. "<< fixedCharges << endl;

cout<< "Meter Rent: Rs. 250\n";
cout<< "TV Fee: Rs. 35\n";

cout<< "GST: Rs. "<< gst << endl;

cout<< "Income Tax: Rs. "<< incomeTax << endl;

cout<<endl<<"Total Payable Bill: Rs. "<<totalBill <<endl;

cout<<"===== "<<endl;
}

// Customer details

void displayCustomerDetails()
{
    cout<<endl<<"===== CUSTOMER DETAILS ===== "<<endl;

    cout<<"Customer Name : "<<customerName<<endl;

```

```
cout<<"Customer ID : "<<customerID<<endl;
```

```
cout<<"Customer Type : "<<customerType<<endl;
```

```
if (meterOption == 1)
```

```
    cout<<"Meter Option : First Meter"<<endl;
```

```
else
```

```
    cout<<"Meter Option : Second Meter"<<endl;
```

```
}
```